Ex: 05

Date: 30/07/26 Experiment on packet capture tool: Wireshark

AIM :-

To understand the feature of wireshark as a Packet capture tool and understand encapsulation of information at various layers of a protocol stack.

Packet Sniffer

* Capture Packets sent and received by the computer.
* Display Protocol information of Captured Pockets
* Works Passively
* Receives copies of network Packets for Analysis

Packet Sniffer Tools :-

* TCP dump - Command line Packet capture tool
* Wire shark - Graphical Packet analyzer.

Description :-

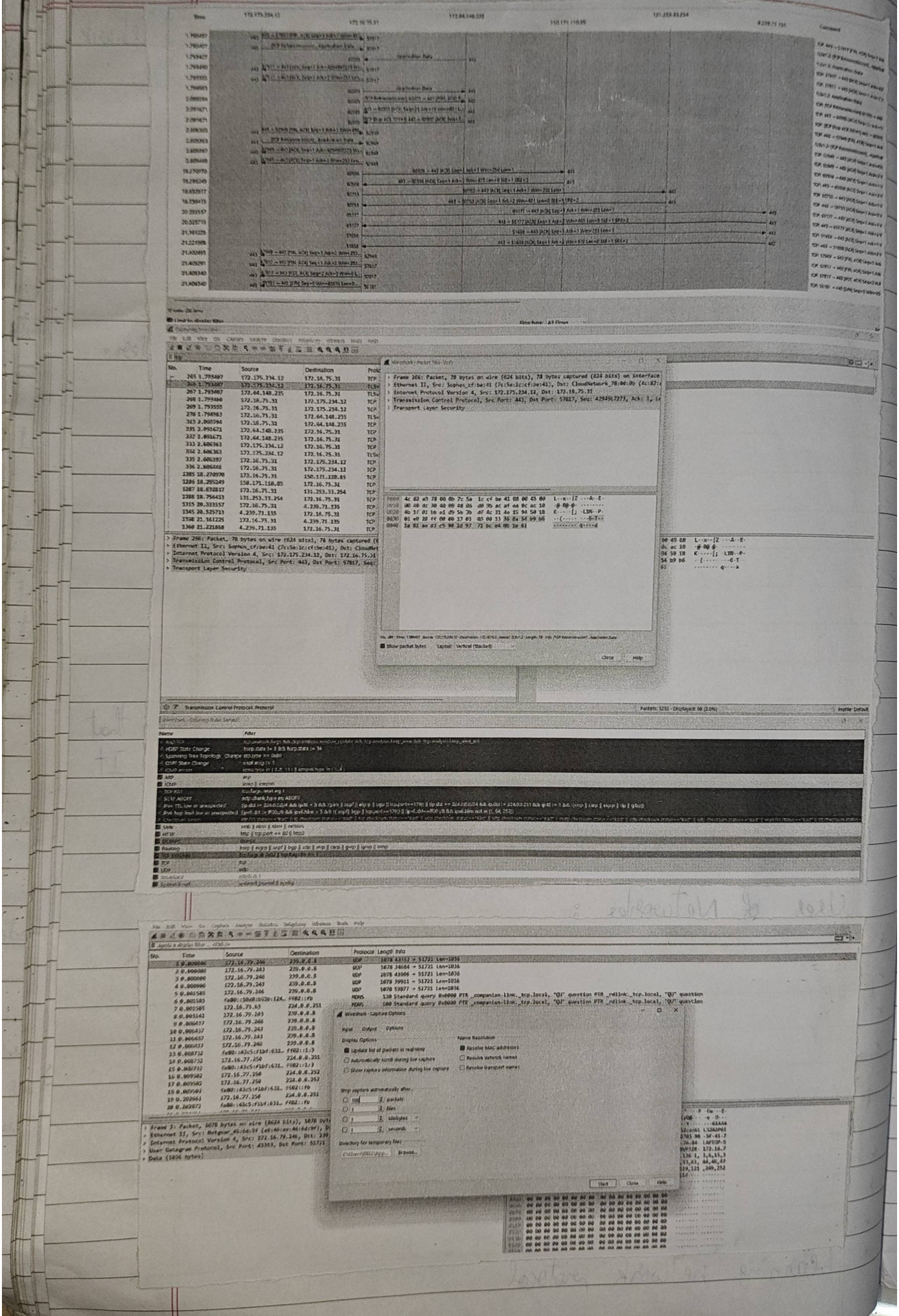
Wireshark is a network protocol analyzer that capture and display network Packets in real time. It helps monitor, analyze, troubleshoot and inspect network traffic using filters and packet details.

Uses of Networks :-

* Capture networks traffic
* Analyzes protocols
* Applies displays filters
* Monitor Network issues

Application :-

* Network Troubleshooting
* Security analysis
* Protocol debugging
* Learning network protocol



Capturing Packet :-

* Open wire shark
* Select the network interface
* Click start Capture
* Observe Packet in real time
* Click stop and save the capture

Color Coding :-

* Purple : TCP
* Blue : UDP
* Black : Error Packets

Filtering Packets :-

* Enter a filter in the filter bar
* Right click to apply filter or follow the TCP stream

Flow Graph :-
Displays the communication flow between devices, making packing, exchange easier to understand.

Procedure :-

1. Create a filter to Display only TCP/UDP Packets.
   (i) Open the Capture file
   (ii) Apply the TCP || UDP filter
   (iii) Inspect and view the flow graph

2.) Create a file to display only ARP Packets
   (i) Open the Capture file
   (ii) Apply the Crop filter
   (iii) Inspect and view the flow graph

3.) Create a filter to display only DNS Packets
(i.) Open the capture file
(ii.) Apply the DNS filter
(iii.) Inspect and view the flow graph.

4.) Create a filter to display only IP/ICMP Packets
(i.) Open the capture file
(ii.) Apply the IP or icmp filter
(iii.) Inspect and view the flow graph and save capture.

5.) Create a filter to display only DHCP Packet
(i.) Open the capture file
(ii.) Apply the dhcp file
(iii.) Inspect and view the flow graph and save the capture.

RESULT :-
Thus the feature of wire shark and the implement of different filter was successfully compiled.